

Seat No.	
----------	--

Set	P
-----	---

**B.Sc. (E.C.S.) (Semester - II) (NEP CBCS) Examination:
October/November - 2025
OOP'S with C++ - II (G06-0201)**

Day & Date: Tuesday, 11-11-2025
Time: 03:00 PM To 04:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose the Correct Alternative. (MCQ)

06

- 1) What is Inheritance in C++?
 - a) Wrapping of data into a single class
 - b) Deriving new classes from existing classes
 - c) Overloading of classes
 - d) Classes with same names

- 2) class X, class Y and class Z are derived from class BASE. This is _____ inheritance.
 - a) Multiple
 - b) Multilevel
 - c) Hierarchical
 - d) Single

- 3) If same message is passed to objects of several different classes and all of those can respond in a different way, what is this feature called?
 - a) Inheritance
 - b) Overloading
 - c) Polymorphism
 - d) Overriding

- 4) By default, all the files in C++ are opened in _____ mode
 - a) Text
 - b) Binary
 - c) ASCII
 - d) VTC

- 5) A base class contain pure virtual function is called as _____.
 - a) Private base class
 - b) Abstract base class
 - c) Virtual function
 - d) None of these

- 6) Which header file is required to use file I/O operations?
 - a) <ifstream>
 - b) <ostream>
 - c) <fstream>
 - d) <iostream>

- Q.2 Answer the following questions. (Any three) 06**
- a) Define Runtime Polymorphism?
 - b) Write the syntax of Derived Class.
 - c) State any four stream state member function.
 - d) Define Exception handling.
 - e) Define Template?
- Q.3 Answer the following questions. (Any two) 06**
- a) Write a program in C++ to implement single inheritance.
 - b) Explain compile time polymorphism with example.
 - c) Explain multiple inheritance with example.
- Q.4 Answer the following questions. (Any two) 06**
- a) Write a program in C++ to read a file and to count the number of vowels in that file.
 - b) Explain try, throw and catch statement with example.
 - c) Write a short note on “virtual base class”.
- Q.5 Answer the following questions. (Any One) 06**
- a) Explain stream classes used in C++? Write a program in C++ to copy one file into another file.
 - b) Define pure virtual function, abstract base class? Explain with example.